MODERN ENTERPRISE NETWORK LAB GUIDE

Cisco Packet Tracer · Step-by-Step Build Guide · 2026 Edition

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Complete beginner-friendly walkthrough to build a production-grade

enterprise cybersecurity network from absolute scratch

Zero Trust Architecture	**DMZ Design**	**VLAN Segmentation**	**ASA Firewall**
HSRP Redundancy	**AAA Authentication**	**SIEM Monitoring**	**Layer 2 Security**

Suitable for absolute beginners · No prior networking experience needed

10 Chapters · 50+ Step-by-Step Instructions · Full Configuration Commands

CHAPTER

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CHAPTER 1

Getting Started

Installing Packet Tracer & understanding the interface

What is Cisco Packet Tracer?

Cisco Packet Tracer is a free network simulation software created by Cisco. It allows you to drag-and-drop virtual routers, switches, firewalls, PCs, and servers onto a canvas and configure them exactly like real hardware — without needing any physical equipment. It is used by millions of students, professionals, and labs worldwide.

> **TIP**
>
> Packet Tracer is completely free when you enroll in the free Cisco Networking Academy course at **netacad.com**. Registration takes under 5 minutes.

Step 1 – Download & Install Packet Tracer

1 Step 1: Create a Free Cisco NetAcad Account

Open your web browser and go to:

```
https://www.netacad.com
```

Click **Sign Up** → Fill in your name, email, and password → Verify your email.

2 Step 2: Enroll in the Free Packet Tracer Course

After logging in, search for **'Introduction to Packet Tracer'** course and enroll. This course is 100% free and unlocks the download.

3 Step 3: Download and Install

Go to the course resources section and download Packet Tracer for your operating system:

OS	Installer File	Recommended Version
Windows 10/11	PacketTracer_*.exe	8.2.x or newer
macOS	PacketTracer_*.dmg	8.2.x or newer
Ubuntu Linux	PacketTracer_*.deb	8.2.x or newer

Run the installer and follow on-screen prompts. Accept all defaults. Installation takes about 3–5 minutes.

4 Step 4: Launch Packet Tracer & Log In

Open Packet Tracer from your desktop. When prompted, log in with your Cisco NetAcad credentials. You need to log in at least once to activate the software.

> **TIP**
> After the first login, you can use Packet Tracer offline.

Understanding the Packet Tracer Interface

When Packet Tracer opens, you will see these main areas:

Area	Location	What It Does
Menu Bar	Top of screen	File, Edit, Options, Help menus
Toolbar	Below menu bar	New, Open, Save, Print buttons
Device Panel	Bottom-left	All devices you can drag onto the canvas
Workspace / Canvas	Centre (large area)	Where you build your network
Connections Panel	Bottom-left tabs	Cable types: copper, fibre, serial
Logical/Physical tabs	Top-left of canvas	Switch between logical and physical view
Simulation Panel	Bottom-right	Packet capture and simulation mode

How to Add Devices to the Canvas

In the bottom-left **Device Panel**, you will see icons grouped by category:

- **Routers** – First icon row. Click to expand. Drag a device onto the canvas.
- **Switches** – Second icon row. Includes Layer 2 and Layer 3 switches.
- **Firewalls / Security** – Includes ASA 5505.
- **End Devices** – PCs, Servers, Laptops.
- **Connections** – Copper Straight-Through, Crossover, Fibre, Serial cables.

> **TIP**
> **To connect two devices:** Click the lightning bolt (Connections) icon in the device panel → Choose cable type → Click Device A → Click Device B. Packet Tracer will show you which ports are available.

How to Open a Device CLI

To configure a router or switch:

- Click once on the device in the canvas.
- A popup window appears.

- Click the **CLI** tab.
- Press **Enter** to get to the command prompt.

> **TIP**
> For servers and PCs, click the device → go to **Desktop** tab → open **IP Configuration** or **Terminal**.

Create & Save Your Project

5 Step 5: Create a New File

Click **File** → **New**. This opens a blank canvas. Immediately save it: **File** → **Save As** → Name it **Enterprise_Lab_2026.pkt**

> **WARNING**
> Save your file every 10–15 minutes as you work. Packet Tracer can occasionally crash and you don't want to lose hours of work.

CHAPTER 2

Network Topology Planning

Understanding the full architecture before we build anything

Why Planning Matters

Before placing a single device, real network engineers always draw the topology, assign IP addresses, and plan VLANs. This prevents mistakes and saves hours of reconfiguration. Read this entire chapter first before starting Chapter 3.

Full Device List

You will need to add these devices to Packet Tracer:

Device Name	Type in Packet Tracer	Role
ISP-Router	Router (e.g. 4331)	Simulates the Internet/ISP
ASA-FW	ASA 5505 Firewall	Perimeter firewall & NAT
CORE1	3560 / 3650 Layer 3 Switch	Primary core switch (active)
CORE2	3560 / 3650 Layer 3 Switch	Secondary core switch (standby)
ACC1	2960 Layer 2 Switch	Access switch – HR/Finance/IT
ACC2	2960 Layer 2 Switch	Access switch – SOC/Developers/Wireless
ACC3	2960 Layer 2 Switch	Access switch – Servers/Management
Web-Server	Generic Server	DMZ Public Web Server
DNS-Public	Generic Server	DMZ Public DNS Server
Mail-Server	Generic Server	DMZ Mail Server
AD-Server	Generic Server	Active Directory
DNS-Internal	Generic Server	Internal DNS
DHCP-Server	Generic Server	DHCP Server
File-Server	Generic Server	File Storage
SIEM-Server	Generic Server	Security Monitoring (SIEM)
Backup-Server	Generic Server	Backup / DR
NTP-Server	Generic Server	Time Synchronisation
AAA-Server	Generic Server	AAA / RADIUS Authentication
Admin-PC	PC	Management Admin PC

Device Name	Type in Packet Tracer	Role
Monitor-PC	PC	SOC Monitoring PC
HR-PC	PC	Sample HR department PC
Finance-PC	PC	Sample Finance department PC
IT-PC	PC	Sample IT department PC
Corporate-AP	Wireless AP (or WRT300N)	Corporate WiFi
Guest-AP	Wireless AP (or WRT300N)	Guest WiFi

Complete VLAN & IP Addressing Plan

This is the master IP plan. Keep this table open as you work through every chapter.

VLAN	Name	Subnet	Gateway (SVI IP)	DHCP?
10	HR	10.10.10.0/24	10.10.10.1	Yes – CORE1
20	FINANCE	10.10.20.0/24	10.10.20.1	Yes – CORE1
30	IT	10.10.30.0/24	10.10.30.1	Yes – CORE1
40	SOC	10.10.40.0/24	10.10.40.1	Yes – CORE1
50	DEVELOPERS	10.10.50.0/24	10.10.50.1	Yes – CORE1
80	GUEST	10.10.80.0/24	10.10.80.1	Yes – CORE1
90	WIRELESS	10.10.90.0/24	10.10.90.1	Yes – CORE1
100	SERVER FARM	10.10.100.0/24	10.10.100.1	No – Static IPs
110	DMZ	10.10.110.0/24	10.10.110.1	No – Static IPs
120	MANAGEMENT	10.10.120.0/24	10.10.120.1	No – Static IPs

Static IP Assignments for All Servers

Server	VLAN	Static IP	Subnet Mask	Gateway
Web-Server	110 (DMZ)	10.10.110.10	255.255.255.0	10.10.110.1
DNS-Public	110 (DMZ)	10.10.110.11	255.255.255.0	10.10.110.1
Mail-Server	110 (DMZ)	10.10.110.12	255.255.255.0	10.10.110.1
AD-Server	100 (Farm)	10.10.100.10	255.255.255.0	10.10.100.1
DNS-Internal	100 (Farm)	10.10.100.11	255.255.255.0	10.10.100.1
DHCP-Server	100 (Farm)	10.10.100.12	255.255.255.0	10.10.100.1
File-Server	100 (Farm)	10.10.100.13	255.255.255.0	10.10.100.1
SIEM-Server	100 (Farm)	10.10.100.14	255.255.255.0	10.10.100.1

Server	VLAN	Static IP	Subnet Mask	Gateway
Backup-Server	100 (Farm)	10.10.100.15	255.255.255.0	10.10.100.1
NTP-Server	100 (Farm)	10.10.100.16	255.255.255.0	10.10.100.1
AAA-Server	120 (Mgmt)	10.10.120.10	255.255.255.0	10.10.120.1
Admin-PC	120 (Mgmt)	10.10.120.20	255.255.255.0	10.10.120.1
Monitor-PC	120 (Mgmt)	10.10.120.21	255.255.255.0	10.10.120.1

Internet Edge Addressing

Link	IP Address	Mask	Note
ISP-Router Fa0/0 (towards ASA)	203.0.113.1	/30	Simulated public IP
ASA-FW Outside interface	203.0.113.2	/30	ASA public-facing IP
ASA-FW Inside interface	10.10.0.1	/24	Connects to CORE1
CORE1 uplink to ASA	10.10.0.2	/24	Layer 3 uplink

TIP

Golden rule: Every device in the same VLAN must be in the same subnet. Every device needing to talk across VLANs goes through the Core switch SVI (gateway).

CHAPTER 3

Building the Internet Edge

ISP Router + Cisco ASA Firewall configuration

Place the Devices

1 **Step 1: Add ISP Router to Canvas**

In Packet Tracer device panel → **Routers** → drag a **4331** (or 1941) onto the top of the canvas. Double-click the label and rename it **ISP-Router**.

2 **Step 2: Add ASA Firewall**

Device panel → **Security** (padlock icon) → drag an **ASA 5505** onto the canvas, just below the ISP Router. Rename it **ASA-FW**.

> **NOTE**
> The ASA 5505 is the only ASA model available in Packet Tracer. It has 8 FastEthernet ports. In real enterprises, ASA 5500-X series are used.

3 **Step 3: Connect ISP Router to ASA Firewall**

Click the **lightning bolt (Connections)** icon → select **Copper Straight-Through** cable → click ISP-Router and pick **GigabitEthernet0/0** → click ASA-FW and pick **Ethernet0/0**.

Configure the ISP Router

4 **Step 4: Open ISP Router CLI**

Click on **ISP-Router** → click the **CLI** tab → press **Enter** to activate.

5 **Step 5: Enter Privileged Mode and Configure**

```
! First, enter privileged exec mode
Router>enable
Router#configure terminal

! Set hostname
```

```
Router(config)#hostname ISP-Router

! Configure the interface facing the ASA firewall
ISP-Router(config)#interface GigabitEthernet0/0
ISP-Router(config-if)#ip address 203.0.113.1 255.255.255.252
ISP-Router(config-if)#no shutdown
ISP-Router(config-if)#description Link-to-ASA-Outside
ISP-Router(config-if)#exit

! Add a default static route pointing toward the ASA
ISP-Router(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.2

! Save configuration
ISP-Router(config)#end
ISP-Router#write memory
```

Configure the ASA Firewall

The ASA firewall has three zones: **Outside** (Internet), **Inside** (Internal LAN), and **DMZ** (Public servers). We configure each interface with a name and security level. Security level 100 = most trusted. Security level 0 = least trusted.

6 Step 6: Open ASA CLI and Enter Configuration Mode

```
! Access the ASA CLI
ciscoasa>enable
Password: (press Enter – blank by default)
ciscoasa#configure terminal

! Set hostname
ciscoasa(config)#hostname ASA-FW
```

7 Step 7: Configure Outside Interface (Ethernet0/0)

```
ASA-FW(config)#interface Ethernet0/0
ASA-FW(config-if)#nameif outside
ASA-FW(config-if)#security-level 0
ASA-FW(config-if)#ip address 203.0.113.2 255.255.255.252
ASA-FW(config-if)#no shutdown
ASA-FW(config-if)#exit
```

8 Step 8: Configure Inside Interface (Ethernet0/1)

```
ASA-FW(config)#interface Ethernet0/1
ASA-FW(config-if)#nameif inside
ASA-FW(config-if)#security-level 100
ASA-FW(config-if)#ip address 10.10.0.1 255.255.255.0
ASA-FW(config-if)#no shutdown
ASA-FW(config-if)#exit
```

9 Step 9: Configure DMZ Interface (Ethernet0/2)

```
ASA-FW(config)#interface Ethernet0/2
ASA-FW(config-if)#nameif dmz
ASA-FW(config-if)#security-level 50
ASA-FW(config-if)#ip address 10.10.110.1 255.255.255.0
ASA-FW(config-if)#no shutdown
ASA-FW(config-if)#exit
```

10 Step 10: Configure NAT – PAT for Internal Users

PAT (Port Address Translation) allows all internal users to share the one public IP when browsing the Internet.

```
! Define an access-list to identify internal traffic
ASA-FW(config)#access-list INSIDE_NAT extended permit ip 10.10.0.0 255.255.0.0
any

! Apply NAT (PAT) – all inside traffic goes out using the outside interface IP
ASA-FW(config)#nat (inside,outside) dynamic interface

! Allow inside-to-outside traffic (higher security can always reach lower)
ASA-FW(config)#access-list OUTSIDE_IN extended permit icmp any any
ASA-FW(config)#access-group OUTSIDE_IN in interface outside
```

11 Step 11: Configure Static NAT for DMZ Servers

Static NAT maps a public IP to each DMZ server so Internet users can reach them.

```
! Static NAT: Public IP 203.0.113.10 → Web Server 10.10.110.10
ASA-FW(config)#nat (dmz,outside) static 203.0.113.10

! Allow HTTP traffic to Web Server from Internet
ASA-FW(config)#access-list OUTSIDE_DMZ extended permit tcp any host 203.0.113.10
eq 80
ASA-FW(config)#access-group OUTSIDE_DMZ in interface outside
```

12 Step 12: Add Default Route on ASA

```
! Default route: send all unknown traffic to ISP Router
ASA-FW(config)#route outside 0.0.0.0 0.0.0.0 203.0.113.1
! Route to internal networks via CORE1
ASA-FW(config)#route inside 10.10.0.0 255.255.0.0 10.10.0.2

ASA-FW(config)#end
ASA-FW#write memory
```

SUCCESS

Verification: From ISP-Router CLI, type: **ping 203.0.113.2** — you should get 5 exclamation marks (!!!!!). This confirms the Internet edge is working.

CHAPTER 4

Core Network — CORE1 & CORE2

Layer 3 switching, VLANs, SVIs, and HSRP redundancy

What is a Layer 3 Switch?

A Layer 3 switch can perform both switching (like a normal switch) AND routing (like a router). This means it can route traffic between different VLANs directly, without needing a separate router. This is called **Inter-VLAN Routing**. CORE1 and CORE2 are the 'brain' of our network — all VLANs connect here.

Place the Core Switches

1 **Step 1: Add CORE1 to Canvas**

Device panel → **Switches** → drag a **Cisco 3650-24PS** (or 3560) onto the canvas. Rename it **CORE1**. Place it in the centre of the canvas.

2 **Step 2: Add CORE2 to Canvas**

Drag another 3650/3560 next to CORE1. Rename it **CORE2**. Connect CORE1 to CORE2 with a **Copper Cross-Over** cable (or use Auto if available) on ports **GigabitEthernet1/0/23** ↔ **GigabitEthernet1/0/23**.

3 **Step 3: Connect ASA Inside to CORE1**

Use a **Copper Straight-Through** cable: **ASA-FW Ethernet0/1** ↔ **CORE1 GigabitEthernet1/0/1**.

Configure CORE1 – VLANs and SVIs

4 **Step 4: Enter CORE1 CLI and Enable IP Routing**

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname CORE1

! Enable Layer 3 routing on this switch
CORE1(config)#ip routing
```

5 **Step 5: Create All VLANs on CORE1**

We create every VLAN that exists in our network on the core switch:

```
CORE1(config)#vlan 10
CORE1(config-vlan)#name HR
CORE1(config-vlan)#exit
CORE1(config)#vlan 20
CORE1(config-vlan)#name FINANCE
CORE1(config-vlan)#exit
CORE1(config)#vlan 30
CORE1(config-vlan)#name IT
CORE1(config-vlan)#exit
CORE1(config)#vlan 40
CORE1(config-vlan)#name SOC
CORE1(config-vlan)#exit
CORE1(config)#vlan 50
CORE1(config-vlan)#name DEVELOPERS
CORE1(config-vlan)#exit
CORE1(config)#vlan 80
CORE1(config-vlan)#name GUEST
CORE1(config-vlan)#exit
CORE1(config)#vlan 90
CORE1(config-vlan)#name WIRELESS
CORE1(config-vlan)#exit
CORE1(config)#vlan 100
CORE1(config-vlan)#name SERVER_FARM
CORE1(config-vlan)#exit
CORE1(config)#vlan 110
CORE1(config-vlan)#name DMZ
CORE1(config-vlan)#exit
CORE1(config)#vlan 120
CORE1(config-vlan)#name MANAGEMENT
CORE1(config-vlan)#exit
```

6 Step 6: Create SVIs (Switch Virtual Interfaces) — the VLAN Gateways

An SVI is a virtual interface that acts as the gateway for each VLAN. Think of it as the 'door' that lets traffic enter and leave each VLAN.

```
! HR VLAN gateway
CORE1(config)#interface vlan 10
```

```
CORE1(config-if)#ip address 10.10.10.1 255.255.255.0
CORE1(config-if)#no shutdown
CORE1(config-if)#description HR_VLAN_Gateway
CORE1(config-if)#exit

! Finance VLAN gateway
CORE1(config)#interface vlan 20
CORE1(config-if)#ip address 10.10.20.1 255.255.255.0
CORE1(config-if)#no shutdown
CORE1(config-if)#exit

! IT VLAN gateway
CORE1(config)#interface vlan 30
CORE1(config-if)#ip address 10.10.30.1 255.255.255.0
CORE1(config-if)#no shutdown
CORE1(config-if)#exit

! SOC VLAN gateway
CORE1(config)#interface vlan 40
CORE1(config-if)#ip address 10.10.40.1 255.255.255.0
CORE1(config-if)#no shutdown
CORE1(config-if)#exit

! Developers VLAN gateway
CORE1(config)#interface vlan 50
CORE1(config-if)#ip address 10.10.50.1 255.255.255.0
CORE1(config-if)#no shutdown
CORE1(config-if)#exit

! Guest VLAN gateway
CORE1(config)#interface vlan 80
CORE1(config-if)#ip address 10.10.80.1 255.255.255.0
CORE1(config-if)#no shutdown
CORE1(config-if)#exit

! Wireless VLAN gateway
CORE1(config)#interface vlan 90
CORE1(config-if)#ip address 10.10.90.1 255.255.255.0
CORE1(config-if)#no shutdown
CORE1(config-if)#exit

! Server Farm VLAN gateway
CORE1(config)#interface vlan 100
```

```
CORE1(config-if)#ip address 10.10.100.1 255.255.255.0
CORE1(config-if)#no shutdown
CORE1(config-if)#exit

! DMZ VLAN gateway
CORE1(config)#interface vlan 110
CORE1(config-if)#ip address 10.10.110.1 255.255.255.0
CORE1(config-if)#no shutdown
CORE1(config-if)#exit

! Management VLAN gateway
CORE1(config)#interface vlan 120
CORE1(config-if)#ip address 10.10.120.1 255.255.255.0
CORE1(config-if)#no shutdown
CORE1(config-if)#exit
```

7 **Step 7: Configure Uplink to ASA Firewall (Layer 3 Routed Port)**

```
! Convert the uplink port to a routed (Layer 3) port
CORE1(config)#interface GigabitEthernet1/0/1
CORE1(config-if)#no switchport
CORE1(config-if)#ip address 10.10.0.2 255.255.255.0
CORE1(config-if)#no shutdown
CORE1(config-if)#description Uplink-to-ASA-FW
CORE1(config-if)#exit

! Default route – send Internet traffic to the ASA
CORE1(config)#ip route 0.0.0.0 0.0.0.0 10.10.0.1

CORE1(config)#end
CORE1#write memory
```

Configure HSRP – Gateway Redundancy

HSRP (Hot Standby Router Protocol) creates a **virtual IP address** shared between CORE1 and CORE2. If CORE1 fails, CORE2 instantly takes over as the active gateway. Users never know a switch failed because they always use the same virtual gateway IP.

> **TIP**
> In HSRP, one switch is **Active** (handles traffic) and the other is **Standby** (monitors and takes over if active fails). We use **priority** to decide which is active. Higher priority = Active.

8 Step 8: Configure HSRP on CORE1 (Active) for All VLANs

We configure HSRP on every VLAN SVI. The virtual IP is the '.1' of the VLAN subnet. Here is the example for VLAN 10 (repeat the pattern for all VLANs 20, 30, 40, 50, 80, 90, 100, 110, 120):

```
CORE1(config)#interface vlan 10
! Group 10 = VLAN 10. Priority 110 = Active (default is 100)
CORE1(config-if)#standby 10 ip 10.10.10.1
CORE1(config-if)#standby 10 priority 110
CORE1(config-if)#standby 10 preempt
CORE1(config-if)#exit

! Repeat for VLAN 20:
CORE1(config)#interface vlan 20
CORE1(config-if)#standby 20 ip 10.10.20.1
CORE1(config-if)#standby 20 priority 110
CORE1(config-if)#standby 20 preempt
CORE1(config-if)#exit
! ... repeat same pattern for VLANs 30,40,50,80,90,100,110,120 ...
```

9 Step 9: Configure CORE2 – Standby HSRP

CORE2 gets the exact same SVI IPs as CORE1, but with lower HSRP priority so it stays Standby. Also create all 10 VLANs on CORE2 (same VLAN commands as CORE1).

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname CORE2
CORE2(config)#ip routing

! Create ALL same VLANs as CORE1 (vlan 10 through 120)
! (use same vlan creation commands from Step 5)

! Create SVIs with same IPs as CORE1
CORE2(config)#interface vlan 10
CORE2(config-if)#ip address 10.10.10.2 255.255.255.0
CORE2(config-if)#no shutdown
! HSRP: same virtual IP, but priority 90 = Standby
CORE2(config-if)#standby 10 ip 10.10.10.1
CORE2(config-if)#standby 10 priority 90
CORE2(config-if)#standby 10 preempt
CORE2(config-if)#exit
```

```
! ... repeat for all other VLANs with priority 90 ...

CORE2(config)#end
CORE2#write memory
```

SUCCESS

Verify HSRP: On CORE1, type: **show standby brief**
You should see CORE1 listed as **Active** and CORE2 as **Standby** for each VLAN.

10 Step 10: Configure DHCP on CORE1

CORE1 will act as the DHCP server for all dynamic VLANs (10, 20, 30, 40, 50, 80, 90). Static VLANs (100, 110, 120) do not get DHCP pools.

```
! Exclude static addresses first (so DHCP doesn't assign them)
CORE1(config)#ip dhcp excluded-address 10.10.10.1 10.10.10.10
CORE1(config)#ip dhcp excluded-address 10.10.20.1 10.10.20.10
CORE1(config)#ip dhcp excluded-address 10.10.30.1 10.10.30.10
CORE1(config)#ip dhcp excluded-address 10.10.40.1 10.10.40.10
CORE1(config)#ip dhcp excluded-address 10.10.50.1 10.10.50.10
CORE1(config)#ip dhcp excluded-address 10.10.80.1 10.10.80.10
CORE1(config)#ip dhcp excluded-address 10.10.90.1 10.10.90.10

! Create DHCP pool for HR (VLAN 10)
CORE1(config)#ip dhcp pool HR_POOL
CORE1(dhcp-config)#network 10.10.10.0 255.255.255.0
CORE1(dhcp-config)#default-router 10.10.10.1
CORE1(dhcp-config)#dns-server 10.10.100.11
CORE1(dhcp-config)#exit

! Create pool for FINANCE (VLAN 20) – repeat pattern for each VLAN
CORE1(config)#ip dhcp pool FINANCE_POOL
CORE1(dhcp-config)#network 10.10.20.0 255.255.255.0
CORE1(dhcp-config)#default-router 10.10.20.1
CORE1(dhcp-config)#dns-server 10.10.100.11
CORE1(dhcp-config)#exit
! ... repeat for IT (30), SOC (40), DEVELOPERS (50), GUEST (80), WIRELESS (90)
...
```

TIP

The pattern for each DHCP pool is identical — just change the VLAN number, pool name, network address, and default-router IP.

CHAPTER 5

Access Layer Switches

ACC1, ACC2, ACC3 — connecting all end devices

What Are Access Layer Switches?

Access switches are what your computers, phones, and servers plug into. They work at Layer 2 only — they switch frames based on MAC addresses. The job of access switches is to: (1) assign ports to VLANs, and (2) trunk all that VLAN traffic up to the core switches.

Place the Access Switches

1 **Step 1: Add ACC1, ACC2, ACC3 to Canvas**

Device panel → **Switches** → drag three **2960-24TT** (or 2960) switches onto the canvas below the core switches. Rename them **ACC1**, **ACC2**, and **ACC3**.

2 **Step 2: Connect Access Switches to Core Switches (Trunk Links)**

Each access switch connects to **both** CORE1 and CORE2 for redundancy. Use **Copper Straight-Through** cables:

From Device	From Port	To Device	To Port
ACC1	GigabitEthernet0/1	CORE1	GigabitEthernet1/0/2
ACC1	GigabitEthernet0/2	CORE2	GigabitEthernet1/0/2
ACC2	GigabitEthernet0/1	CORE1	GigabitEthernet1/0/3
ACC2	GigabitEthernet0/2	CORE2	GigabitEthernet1/0/3
ACC3	GigabitEthernet0/1	CORE1	GigabitEthernet1/0/4
ACC3	GigabitEthernet0/2	CORE2	GigabitEthernet1/0/4

Configure ACC1 – HR, Finance, IT VLANs

3 **Step 3: Configure ACC1 VLANs and Trunk Ports**

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname ACC1
```

```
! Create VLANs on this switch
ACC1(config)#vlan 10
ACC1(config-vlan)#name HR
ACC1(config-vlan)#exit
ACC1(config)#vlan 20
ACC1(config-vlan)#name FINANCE
ACC1(config-vlan)#exit
ACC1(config)#vlan 30
ACC1(config-vlan)#name IT
ACC1(config-vlan)#exit

! Configure trunk port to CORE1 (GigabitEthernet0/1)
ACC1(config)#interface GigabitEthernet0/1
ACC1(config-if)#switchport mode trunk
ACC1(config-if)#switchport trunk allowed vlan 10,20,30
ACC1(config-if)#description Trunk-to-CORE1
ACC1(config-if)#no shutdown
ACC1(config-if)#exit

! Configure trunk port to CORE2 (GigabitEthernet0/2)
ACC1(config)#interface GigabitEthernet0/2
ACC1(config-if)#switchport mode trunk
ACC1(config-if)#switchport trunk allowed vlan 10,20,30
ACC1(config-if)#description Trunk-to-CORE2
ACC1(config-if)#no shutdown
ACC1(config-if)#exit
```

4 Step 4: Configure Access Ports on ACC1 (HR PC, Finance PC, IT PC)

Access ports connect end devices (PCs). Each port is assigned to ONE VLAN.

```
! HR-PC connects to FastEthernet0/1 → VLAN 10
ACC1(config)#interface FastEthernet0/1
ACC1(config-if)#switchport mode access
ACC1(config-if)#switchport access vlan 10
ACC1(config-if)#description HR-PC
ACC1(config-if)#spanning-tree portfast
ACC1(config-if)#no shutdown
ACC1(config-if)#exit

! Finance-PC connects to FastEthernet0/2 → VLAN 20
```

```
ACC1(config)#interface FastEthernet0/2
ACC1(config-if)#switchport mode access
ACC1(config-if)#switchport access vlan 20
ACC1(config-if)#description Finance-PC
ACC1(config-if)#spanning-tree portfast
ACC1(config-if)#no shutdown
ACC1(config-if)#exit

! IT-PC connects to FastEthernet0/3 → VLAN 30
ACC1(config)#interface FastEthernet0/3
ACC1(config-if)#switchport mode access
ACC1(config-if)#switchport access vlan 30
ACC1(config-if)#description IT-PC
ACC1(config-if)#spanning-tree portfast
ACC1(config-if)#no shutdown
ACC1(config-if)#exit

ACC1(config)#end
ACC1#write memory
```

5 Step 5: Configure ACC2 – SOC, Developers, Wireless VLANs

Follow the exact same process for ACC2. Create VLANs 40, 50, 90 on it, set trunk ports to CORE1/CORE2, and set access ports for SOC-PC, Developer-PC, and Wireless AP.

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname ACC2
ACC2(config)#vlan 40
ACC2(config-vlan)#name SOC
ACC2(config-vlan)#exit
ACC2(config)#vlan 50
ACC2(config-vlan)#name DEVELOPERS
ACC2(config-vlan)#exit
ACC2(config)#vlan 80
ACC2(config-vlan)#name GUEST
ACC2(config-vlan)#exit
ACC2(config)#vlan 90
ACC2(config-vlan)#name WIRELESS
ACC2(config-vlan)#exit
```

```
! Trunk ports to CORE1 and CORE2 (same as ACC1 but allow VLANs 40,50,80,90)
ACC2(config)#interface GigabitEthernet0/1
ACC2(config-if)#switchport mode trunk
ACC2(config-if)#switchport trunk allowed vlan 40,50,80,90
ACC2(config-if)#no shutdown
ACC2(config-if)#exit
ACC2(config)#interface GigabitEthernet0/2
ACC2(config-if)#switchport mode trunk
ACC2(config-if)#switchport trunk allowed vlan 40,50,80,90
ACC2(config-if)#no shutdown
ACC2(config-if)#exit

! Access port for Wireless AP (VLAN 90)
ACC2(config)#interface FastEthernet0/1
ACC2(config-if)#switchport mode access
ACC2(config-if)#switchport access vlan 90
ACC2(config-if)#description Corporate-AP
ACC2(config-if)#spanning-tree portfast
ACC2(config-if)#exit

! Guest AP access port (VLAN 80)
ACC2(config)#interface FastEthernet0/2
ACC2(config-if)#switchport mode access
ACC2(config-if)#switchport access vlan 80
ACC2(config-if)#description Guest-AP
ACC2(config-if)#spanning-tree portfast
ACC2(config-if)#exit

ACC2(config)#end
ACC2#write memory
```

6 Step 6: Configure ACC3 – Server Farm and Management VLANs

ACC3 connects all servers and management devices. Trunk VLANs 100 and 120 to core switches.

```
Switch>enable
Switch#configure terminal
Switch(config)#hostname ACC3
ACC3(config)#vlan 100
ACC3(config-vlan)#name SERVER_FARM
ACC3(config-vlan)#exit
```

```
ACC3(config)#vlan 120
ACC3(config-vlan)#name MANAGEMENT
ACC3(config-vlan)#exit

! Trunk to CORE1 and CORE2
ACC3(config)#interface GigabitEthernet0/1
ACC3(config-if)#switchport mode trunk
ACC3(config-if)#switchport trunk allowed vlan 100,110,120
ACC3(config-if)#no shutdown
ACC3(config-if)#exit
ACC3(config)#interface GigabitEthernet0/2
ACC3(config-if)#switchport mode trunk
ACC3(config-if)#switchport trunk allowed vlan 100,110,120
ACC3(config-if)#no shutdown
ACC3(config-if)#exit

! Each server gets its own access port (Fa0/1=AD, Fa0/2=DNS, etc.)
ACC3(config)#interface FastEthernet0/1
ACC3(config-if)#switchport mode access
ACC3(config-if)#switchport access vlan 100
ACC3(config-if)#description AD-Server
ACC3(config-if)#spanning-tree portfast
ACC3(config-if)#exit
! Repeat same pattern for DNS(Fa0/2), DHCP(Fa0/3), File(Fa0/4),
! SIEM(Fa0/5), Backup(Fa0/6), NTP(Fa0/7) all in VLAN 100

! Management ports – VLAN 120
ACC3(config)#interface FastEthernet0/20
ACC3(config-if)#switchport mode access
ACC3(config-if)#switchport access vlan 120
ACC3(config-if)#description AAA-Server
ACC3(config-if)#exit
ACC3(config)#interface FastEthernet0/21
ACC3(config-if)#switchport mode access
ACC3(config-if)#switchport access vlan 120
ACC3(config-if)#description Admin-PC
ACC3(config-if)#exit

ACC3(config)#end
ACC3#write memory
```

7 Step 7: Also Configure Trunk Ports on CORE1 and CORE2 Facing Access Switches

The ports on CORE1 and CORE2 facing access switches must also be configured as trunks:

```
! On CORE1:
CORE1(config)#interface GigabitEthernet1/0/2
CORE1(config-if)#switchport trunk encapsulation dot1q
CORE1(config-if)#switchport mode trunk
CORE1(config-if)#switchport trunk allowed vlan 10,20,30
CORE1(config-if)#description Trunk-to-ACC1
CORE1(config-if)#no shutdown
CORE1(config-if)#exit
! Repeat for GigabitEthernet1/0/3 (to ACC2) and 1/0/4 (to ACC3)
! with appropriate VLAN lists for each
```

WARNING

Layer 3 switches need 'switchport trunk encapsulation dot1q' before 'switchport mode trunk'. Layer 2 (2960) switches don't need this command.

Assign Static IPs to All PCs and Servers

For PCs that use DHCP: Click the PC → Desktop tab → IP Configuration → select **DHCP**. The PC will automatically get an IP from CORE1.

For servers and management devices with static IPs: Click the server → Desktop tab → IP Configuration → select **Static** → enter the IP, subnet mask, and default gateway from the table in Chapter 2.

CHAPTER 6

DMZ Architecture

Isolating public-facing servers from internal networks

What is a DMZ and Why Do We Need It?

DMZ stands for **De-Militarized Zone**. It is a network segment that sits **between** the Internet and your internal network. Public servers (web, DNS, mail) are placed in the DMZ. If an attacker hacks your web server, they are stuck in the DMZ — they **cannot** reach your internal servers like Active Directory or the File Server. The firewall blocks DMZ-to-Internal traffic.

WARNING

Without a DMZ, a hacked web server gives attackers direct access to your entire internal network. The DMZ is one of the most important security controls in enterprise design.

Place DMZ Servers

1 Step 1: Add DMZ Servers to Canvas

Drag three **Generic Servers** from the end-devices panel onto the canvas. Place them near the ASA firewall. Rename them:

- **Web-Server** — hosts public website
- **DNS-Public** — public DNS resolution
- **Mail-Server** — email handling

2 Step 2: Connect DMZ Servers to ASA Firewall

In a real network, DMZ servers connect through a switch. In Packet Tracer, to keep it simple, connect directly to ASA Ethernet0/2 (the DMZ interface). Use a small switch (drag a 2960) as a DMZ switch and connect all three servers to it, then connect the DMZ switch to ASA Ethernet0/2.

```
! DMZ switch config (rename a 2960 to 'DMZ-SW')
Switch(config)#hostname DMZ-SW
DMZ-SW(config)#vlan 110
DMZ-SW(config-vlan)#name DMZ
DMZ-SW(config-vlan)#exit

! All server ports as access VLAN 110
DMZ-SW(config)#interface FastEthernet0/1
```

```
DMZ-SW(config-if)#switchport mode access
DMZ-SW(config-if)#switchport access vlan 110
DMZ-SW(config-if)#description Web-Server
DMZ-SW(config-if)#exit
! Repeat for Fa0/2 (DNS-Public) and Fa0/3 (Mail-Server)
DMZ-SW(config)#end
DMZ-SW#write memory
```

3 Step 3: Set Static IPs on DMZ Servers

Click each DMZ server → **Desktop** tab → **IP Configuration** → Static:

Server	IP Address	Subnet Mask	Default Gateway
Web-Server	10.10.110.10	255.255.255.0	10.10.110.1
DNS-Public	10.10.110.11	255.255.255.0	10.10.110.1
Mail-Server	10.10.110.12	255.255.255.0	10.10.110.1

TIP

The gateway for DMZ servers is **10.10.110.1** which is the ASA's DMZ interface — NOT the CORE switch. DMZ traffic goes directly through the firewall.

Configure Web Server Services

4 Step 4: Enable HTTP on Web-Server

Click **Web-Server** → go to the **Services** tab → click **HTTP** → make sure it is **ON**. You can edit the default HTML page content here. This simulates a public website.

5 Step 5: Enable DNS on DNS-Public Server

Click **DNS-Public** → **Services** tab → click **DNS** → turn it **ON**.

Add a DNS record: Name = **www.enterprise.com** → Address = **10.10.110.10**

ASA Firewall – DMZ Security Rules

The ASA by default allows higher-security zones to reach lower-security zones. But we must explicitly **block** DMZ-to-Internal traffic to prevent lateral movement if a DMZ server is compromised.

6 Step 6: Block DMZ-to-Internal Access on ASA

```
! Block DMZ from reaching internal network
```

```
ASA-FW(config)#access-list DMZ_IN extended deny ip 10.10.110.0 255.255.255.0
10.10.0.0 255.255.0.0

! Allow DMZ to reach Internet (for updates, etc.)

ASA-FW(config)#access-list DMZ_IN extended permit ip 10.10.110.0 255.255.255.0
any

! Apply the rule to DMZ interface incoming traffic

ASA-FW(config)#access-group DMZ_IN in interface dmz

ASA-FW(config)#end

ASA-FW#write memory
```

CHAPTER 7

Internal Server Farm

Active Directory, DNS, DHCP, SIEM, Backup, NTP — all critical internal services

Overview of Internal Servers

The internal server farm lives on VLAN 100 (10.10.100.0/24) and is connected via ACC3. Each server provides a specific service. Here is a summary:

Server	IP	Service Purpose	Packet Tracer Service Tab
AD-Server	10.10.100.10	User authentication & directory services	N/A (simulated)
DNS-Internal	10.10.100.11	Resolves hostnames for internal systems	Services → DNS
DHCP-Server	10.10.100.12	Auto assigns IPs (we use CORE1 instead)	Services → DHCP
File-Server	10.10.100.13	Central file storage	Services → FTP or HTTP
SIEM-Server	10.10.100.14	Receives Syslog logs, monitors events	Services → Syslog
Backup-Server	10.10.100.15	Data backup and disaster recovery	Services → FTP
NTP-Server	10.10.100.16	Time synchronization for all devices	Services → NTP

Setting Up Server IPs

1 **Step 1: Assign Static IPs to All Internal Servers**

For **each server** on ACC3: Click the server → **Desktop** tab → **IP Configuration** → select **Static.** Enter IP, mask, and gateway from the table above.

WARNING

All servers are on VLAN 100. Their gateway is **10.10.100.1** (CORE1 SVI). Never use DHCP for servers — static IPs ensure consistent reachability.

Configure DNS-Internal Server

2 **Step 2: Enable and Configure DNS**

Click **DNS-Internal** → **Services** tab → **DNS** → turn **ON**.

Add these DNS records (click **Add** for each):

Name	Type	Address
ad.enterprise.local	A Record	10.10.100.10
dns.enterprise.local	A Record	10.10.100.11
dhcp.enterprise.local	A Record	10.10.100.12
files.enterprise.local	A Record	10.10.100.13
siem.enterprise.local	A Record	10.10.100.14
ntp.enterprise.local	A Record	10.10.100.16

Configure NTP Server

3 **Step 3: Enable NTP Service**

Click **NTP-Server** → **Services** tab → **NTP** → turn **ON**. The NTP server will now provide time to all network devices.

4 **Step 4: Configure All Network Devices to Use NTP**

On each router and switch (CORE1, CORE2, ACC1, ACC2, ACC3, ASA), add this command:

```
! Point all devices to the NTP server
(config)#ntp server 10.10.100.16
```

Configure SIEM / Syslog Server

5 **Step 5: Enable Syslog on SIEM-Server**

Click **SIEM-Server** → **Services** tab → **Syslog** → turn **ON**. This server will now collect log messages from all network devices.

6 **Step 6: Configure All Devices to Send Logs to SIEM**

On each router and switch, configure syslog logging:

```
! Enable logging and point to SIEM server
(config)#logging 10.10.100.14
(config)#logging on
```

```
(config)#logging trap informational
(config)#service timestamps log datetime msec
```

Configure SNMP Monitoring

7 Step 7: Enable SNMP on All Switches

```
! Configure SNMP community string for monitoring
(config)#snmp-server community EnterpriseRONLY ro
(config)#snmp-server location Enterprise-DataCenter
(config)#snmp-server contact admin@enterprise.local
! Send SNMP traps to SIEM server
(config)#snmp-server host 10.10.100.14 version 2c EnterpriseRONLY
```

TIP

SNMP 'ro' means read-only. This is safer — the monitoring server can READ device stats but cannot make any configuration changes.

CHAPTER 8

Security Hardening

SSH, Port Security, DHCP Snooping, DAI, BPDU Guard, AAA

Why Hardening Matters

Configuring devices is not enough — we must also harden them against attacks. Every unprotected interface, weak password, or open service is an opportunity for an attacker. This chapter applies critical Layer 2 and management-plane security controls to every switch and router in the lab.

SSH — Secure Remote Management

By default, Cisco devices allow Telnet — which sends passwords in plaintext. We replace Telnet with SSH which encrypts all management traffic.

1 Step 1: Configure SSH on ALL Switches and Routers

Apply this configuration to **every device**: CORE1, CORE2, ACC1, ACC2, ACC3, ISP-Router.

```
! Set a domain name (required for RSA key generation)
(config)#ip domain-name enterprise.local

! Generate the RSA encryption keys
(config)#crypto key generate rsa modulus 2048
! (Press Enter when prompted)

! Create a local admin user
(config)#username admin privilege 15 secret Cisco@12345!

! Configure SSH version 2 only (more secure)
(config)#ip ssh version 2
(config)#ip ssh time-out 60
(config)#ip ssh authentication-retries 3

! Apply SSH to VTY lines (remote management lines)
(config)#line vty 0 4
(config-line)#transport input ssh
(config-line)#login local
(config-line)#exec-timeout 10 0
(config-line)#exit
```

```
! Secure the console port
(config)#line console 0
(config-line)#login local
(config-line)#exec-timeout 10 0
(config-line)#exit

! Enable password encryption in config file
(config)#service password-encryption
(config)#enable secret Cisco@Enable99!
```

Port Security

Port security limits which MAC addresses can connect to each switch port. If an unknown device plugs in, the port shuts down automatically.

2 Step 2: Configure Port Security on All Access Ports

Apply to every **access port** on ACC1, ACC2, ACC3 (not trunk ports):

```
! Example: FastEthernet0/1 on ACC1 (HR-PC port)
ACC1(config)#interface FastEthernet0/1
ACC1(config-if)#switchport port-security
! Allow maximum 1 MAC address
ACC1(config-if)#switchport port-security maximum 1
! If violation: shut the port down
ACC1(config-if)#switchport port-security violation shutdown
! Learn the current MAC automatically and save it
ACC1(config-if)#switchport port-security mac-address sticky
ACC1(config-if)#exit
! Apply same config to all other access ports (Fa0/2, Fa0/3, etc.)
```

TIP

If a port shuts down due to a security violation, re-enable it with: **interface Fa0/X** → **shutdown** → **no shutdown**

DHCP Snooping

DHCP Snooping prevents rogue DHCP servers. An attacker could plug in a device offering fake DHCP replies with their own IP as the gateway — this is a Man-in-the-Middle attack. DHCP Snooping only allows DHCP replies from trusted ports (uplinks to core switches).

3 Step 3: Configure DHCP Snooping on ALL Access Switches

Apply to ACC1, ACC2, ACC3:

```
! Enable DHCP snooping globally
(config)#ip dhcp snooping

! Enable for all VLANs on this switch
(config)#ip dhcp snooping vlan 10,20,30,40,50,80,90,100,110,120

! Mark the uplink port to CORE1 as TRUSTED
! (Only trusted ports can send DHCP replies)
(config)#interface GigabitEthernet0/1
(config-if)#ip dhcp snooping trust
(config-if)#exit

! The uplink to CORE2 is also trusted
(config)#interface GigabitEthernet0/2
(config-if)#ip dhcp snooping trust
(config-if)#exit

! All other (access) ports remain UNTRUSTED by default
```

Dynamic ARP Inspection (DAI)

ARP spoofing is when an attacker sends fake ARP replies to poison ARP tables, making traffic go to the attacker instead of the real gateway. DAI validates every ARP packet against the DHCP snooping database. Fake ARP packets are dropped.

4 Step 4: Configure DAI on ALL Access Switches

```
! Enable DAI for all VLANs
(config)#ip arp inspection vlan 10,20,30,40,50,80,90,100,110,120

! Trust the uplink ports (same as DHCP snooping trusted ports)
(config)#interface GigabitEthernet0/1
(config-if)#ip arp inspection trust
(config-if)#exit
(config)#interface GigabitEthernet0/2
(config-if)#ip arp inspection trust
(config-if)#exit
```

BPDU Guard

STP (Spanning Tree Protocol) prevents loops in Ethernet networks. BPDU Guard protects against someone connecting a rogue switch to an access port. If any BPDU (Spanning Tree message) arrives on an edge port, the port is shut down immediately.

5 Step 5: Enable BPDU Guard on All Access Ports

```
! Enable PortFast and BPDU Guard globally on all access ports
(config)#spanning-tree portfast default
(config)#spanning-tree portfast bpduguard default

! Or configure per-interface:
(config)#interface FastEthernet0/1
(config-if)#spanning-tree portfast
(config-if)#spanning-tree bpduguard enable
(config-if)#exit
```

Disable Unused Ports

6 Step 6: Shut Down All Unused Switch Ports

Any unused port is a security risk — an attacker could plug in undetected. Shut down all ports that have nothing connected.

```
! Example: Shut down ports Fa0/10 through Fa0/24 on ACC1
ACC1(config)#interface range FastEthernet0/10 - 24
ACC1(config-if-range)#shutdown
ACC1(config-if-range)#description UNUSED_DISABLED
ACC1(config-if-range)#exit
```

AAA Authentication

AAA = Authentication (who are you?), Authorization (what can you do?), Accounting (what did you do?). We configure TACACS+ pointing to the AAA-Server so that every admin login is centrally controlled and logged.

7 Step 7: Configure AAA on All Switches and Routers

```
! Enable AAA globally
(config)#aaa new-model

! Define the TACACS+ server (AAA-Server at 10.10.120.10)
(config)#tacacs server ENTERPRISE_AAA
(config-server-tacacs)#address ipv4 10.10.120.10
(config-server-tacacs)#key EnterpriseAAA@Key!
(config-server-tacacs)#exit

! Create AAA authentication group using the TACACS+ server
```

```
(config)#aaa group server tacacs+ AAA_GROUP
(config-sg-tacacs+)#server name ENTERPRISE_AAA
(config-sg-tacacs+)#exit

! Use AAA group for login, fall back to local if server unreachable
(config)#aaa authentication login default group AAA_GROUP local
(config)#aaa authorization exec default group AAA_GROUP local
(config)#aaa accounting exec default start-stop group AAA_GROUP
```

NOTE

In Packet Tracer, full TACACS+ functionality is limited. The commands above are correct for real Cisco devices. In PT, the local username/password will be used as fallback and the AAA authentication concepts are demonstrated.

CHAPTER 9

ACL Segmentation & Zero Trust

Enforcing least privilege with Access Control Lists

What is an ACL?

An ACL (Access Control List) is a set of rules that tells a router or switch which traffic to allow or block. ACLs are our primary tool for enforcing **Zero Trust** — the principle that no device is automatically trusted just because it is on the network. Every flow must be explicitly permitted.

TIP

ACLs are applied to **interfaces** in a specific direction: **in** (traffic entering the interface) or **out** (traffic leaving). On CORE1, we apply ACLs to VLAN SVIs to filter inter-VLAN traffic.

ACL Design Summary

VLAN / Zone	Blocked From	Allowed To	Reason
GUEST (80)	All internal networks	Internet only	Visitor isolation
WIRELESS (90)	Management VLAN (120)	Other VLANs + Internet	Wireless separation
HR (10)	Finance VLAN (20)	IT, Servers, Internet	Department isolation
DMZ (110)	All internal (10.10.0.0/16)	Internet	DMZ protection
SOC (40)	Nothing blocked	All including SIEM	SOC needs full visibility
IT (30)	Nothing blocked	All including Management	IT admins need access

Configure ACLs on CORE1

1 Step 1: Guest VLAN Isolation (VLAN 80)

Guests should only reach the Internet. They must never access internal systems.

```
! Create ACL 100 for Guest VLAN
CORE1(config)#ip access-list extended GUEST_RESTRICT
! Deny all traffic from Guest to any internal 10.10.x.x address
CORE1(config-ext-nacl)#deny ip 10.10.80.0 0.0.0.255 10.10.0.0 0.0.255.255
! Allow Guest to reach Internet (any other destination)
```

```
CORE1(config-ext-nacl)#permit ip 10.10.80.0 0.0.0.255 any
CORE1(config-ext-nacl)#exit

! Apply this ACL outbound on VLAN 80 SVI
CORE1(config)#interface vlan 80
CORE1(config-if)#ip access-group GUEST_RESTRICT in
CORE1(config-if)#exit
```

2 Step 2: Wireless VLAN Isolation (VLAN 90)

Wireless devices cannot reach the Management VLAN.

```
CORE1(config)#ip access-list extended WIRELESS_RESTRICT
! Block Wireless from reaching Management VLAN
CORE1(config-ext-nacl)#deny ip 10.10.90.0 0.0.0.255 10.10.120.0 0.0.0.255
! Permit everything else
CORE1(config-ext-nacl)#permit ip 10.10.90.0 0.0.0.255 any
CORE1(config-ext-nacl)#exit

CORE1(config)#interface vlan 90
CORE1(config-if)#ip access-group WIRELESS_RESTRICT in
CORE1(config-if)#exit
```

3 Step 3: HR Cannot Access Finance (Department Separation)

```
CORE1(config)#ip access-list extended HR_RESTRICT
! HR cannot access Finance VLAN
CORE1(config-ext-nacl)#deny ip 10.10.10.0 0.0.0.255 10.10.20.0 0.0.0.255
! HR can access servers and everything else
CORE1(config-ext-nacl)#permit ip 10.10.10.0 0.0.0.255 any
CORE1(config-ext-nacl)#exit

CORE1(config)#interface vlan 10
CORE1(config-if)#ip access-group HR_RESTRICT in
CORE1(config-if)#exit
```

4 Step 4: SOC Full Access (SOC Needs to See Everything)

```
CORE1(config)#ip access-list extended SOC_ACCESS
! SOC needs to reach SIEM server and all VLANs for monitoring
CORE1(config-ext-nacl)#permit ip 10.10.40.0 0.0.0.255 10.10.100.14 0.0.0.0
CORE1(config-ext-nacl)#permit ip 10.10.40.0 0.0.0.255 any
```

```
CORE1(config-ext-nacl)#exit

CORE1(config)#interface vlan 40
CORE1(config-if)#ip access-group SOC_ACCESS in
CORE1(config-if)#exit
```

5 Step 5: Management VLAN – Only IT Admins Can Access

Only the IT VLAN and Management VLAN itself should reach the Management VLAN.

```
CORE1(config)#ip access-list extended MGMT_PROTECT
! Allow IT VLAN (admins) to reach Management
CORE1(config-ext-nacl)#permit ip 10.10.30.0 0.0.0.255 10.10.120.0 0.0.0.255
! Allow Management VLAN itself
CORE1(config-ext-nacl)#permit ip 10.10.120.0 0.0.0.255 10.10.120.0 0.0.0.255
! Deny everything else from reaching Management
CORE1(config-ext-nacl)#deny ip any 10.10.120.0 0.0.0.255
! Permit all other traffic (not aimed at Management)
CORE1(config-ext-nacl)#permit ip any any
CORE1(config-ext-nacl)#exit

CORE1(config)#interface vlan 120
CORE1(config-if)#ip access-group MGMT_PROTECT in
CORE1(config-if)#exit

CORE1(config)#end
CORE1#write memory
```

WARNING

Important: ACLs are processed top-to-bottom. The first matching rule wins. Always put specific rules before general ones. A final **deny any any** is implicit — traffic not matched by any permit is dropped.

CHAPTER 10

Validation & Testing

Verifying every component works correctly

How to Test in Packet Tracer

Packet Tracer has two testing methods: (1) the **ping** command typed in device CLIs, and (2) the **Add Simple PDU** button (envelope icon in the toolbar) to visually send test packets. Use CLI ping for most tests — it gives you clear pass/fail output.

Test 1 – Basic Connectivity

1 **Step 1: Test PC-to-Gateway Connectivity**

From HR-PC (should have DHCP IP in 10.10.10.x), open the PC → Desktop → Command Prompt:

```
C:\> ping 10.10.10.1
! Expected: 5 reply packets (!!!!!)
! If failed: check VLAN assignment, DHCP, and SVI config on CORE1
```

2 **Step 2: Test Inter-VLAN Routing**

From HR-PC, ping the File Server on VLAN 100:

```
C:\> ping 10.10.100.13
! Expected: Success (5 replies)
! This confirms inter-VLAN routing on CORE1 is working
```

3 **Step 3: Test Internet Connectivity**

From HR-PC, ping the ISP Router:

```
C:\> ping 203.0.113.1
! Expected: Success – confirms NAT/PAT and routing through ASA is working
```

Test 2 – ACL Verification

4 **Step 4: Test Guest Isolation**

From a Guest VLAN PC (IP in 10.10.80.x), try to reach an internal server:

```
C:\> ping 10.10.100.10
! Expected: FAIL (timeout or request timed out)
! This confirms the Guest ACL is blocking internal access

C:\> ping 203.0.113.1
! Expected: SUCCESS – Guest can still reach the Internet
```

5 Step 5: Test HR Cannot Reach Finance

From HR-PC, try to ping Finance-PC (10.10.20.x):

```
C:\> ping 10.10.20.50
! Expected: FAIL – HR ACL blocks Finance VLAN access
```

6 Step 6: Test Wireless Cannot Reach Management

From a Wireless VLAN PC (10.10.90.x), try to reach the Admin-PC (10.10.120.20):

```
C:\> ping 10.10.120.20
! Expected: FAIL – Wireless ACL blocks Management VLAN
```

Test 3 – HSRP Failover

7 Step 7: Verify HSRP Status

On CORE1 CLI:

```
CORE1#show standby brief
! Expected output shows CORE1 as Active for all VLANs
! CORE2 should show as Standby
```

8 Step 8: Simulate CORE1 Failure

Click on **CORE1** → go to the **Physical** tab → click the power button to turn off. Wait 10–15 seconds. Then from HR-PC:

```
C:\> ping 10.10.100.13
! After brief pause, pings should resume
! HSRP failover: CORE2 became Active

! Check on CORE2:
CORE2#show standby brief
! CORE2 now shows as Active for all VLANs
```

Turn CORE1 back on. After a few seconds, CORE1 will reclaim the Active role (because of **preempt** and higher priority).

Test 4 – SSH Verification

9 **Step 9: Test SSH Access to a Switch**

From Admin-PC (10.10.120.20), open Desktop → Terminal (or Command Prompt):

```
C:\> ssh -l admin 10.10.120.1
! Enter password: Cisco@12345!
! Expected: You should get a CORE1 command prompt
! This confirms SSH management is working
```

Test 5 – DHCP Verification

10 **Step 10: Verify DHCP Leases on CORE1**

```
CORE1#show ip dhcp binding
! Shows all IPs assigned via DHCP with MAC addresses
CORE1#show ip dhcp pool
! Shows pool utilization
```

Final Validation Checklist

Go through this checklist before calling the lab complete:

- ✓ All PCs in dynamic VLANs (10,20,30,40,50,80,90) received DHCP IPs
- ✓ All servers have correct static IPs assigned
- ✓ Inter-VLAN routing works (HR-PC can ping File-Server)
- ✓ Internet connectivity works (HR-PC can ping ISP-Router)
- ✓ Guest VLAN cannot reach any internal IP
- ✓ Wireless VLAN cannot reach Management VLAN
- ✓ HR VLAN cannot reach Finance VLAN
- ✓ DMZ servers are reachable from outside but cannot reach internal VLANs
- ✓ HSRP failover tested – network survives CORE1 shutdown
- ✓ SSH access working from Admin-PC to switches
- ✓ DHCP Snooping enabled on all access switches
- ✓ DAI enabled on all access switches
- ✓ BPDU Guard enabled on all access ports
- ✓ Port security enabled on access ports
- ✓ Unused ports shut down on all switches

✓ Syslog configured pointing to SIEM-Server (10.10.100.14)

✓ NTP configured on all devices pointing to 10.10.100.16

✓ All device configurations saved with 'write memory'

SUCCESS

Congratulations! If all checklist items pass, you have successfully built a production-grade enterprise network from scratch in Cisco Packet Tracer. This architecture reflects the same principles used in real enterprise environments in banking, healthcare, government, and technology companies. You have implemented: Zero Trust, Defense in Depth, High Availability, Layered Security, SOC Visibility, and Enterprise Segmentation.

Quick Reference — Most Useful Show Commands

Command	Device	What It Shows
show ip interface brief	Any	All interfaces, IPs, and up/down status
show vlan brief	Switch	All VLANs and port assignments
show interfaces trunk	Switch	Trunk ports and allowed VLANs
show standby brief	CORE1/2	HSRP active/standby status
show ip dhcp binding	CORE1	All DHCP-assigned IPs
show ip route	Any	Routing table
show access-lists	CORE1	All ACLs and match counts
show port-security interface	Switch	Port security status per port
show ip dhcp snooping binding	Switch	DHCP snooping binding table
show ip arp inspection	Switch	DAI statistics
show ssh	Any	Active SSH sessions
show running-config	Any	Full current configuration

WARNING

Always save your configuration with **write memory** (or **copy running-config startup-config**) after making any changes. Unsaved configs are lost when Packet Tracer closes.